PIPELINE AUDIT AND HEALTH REPORT

Bike assembly: custom nodes showcase

Two part streams meet in an assembler (1 frame + 2 wheels). A custom 'quality_gate' node inspects each bike, sends defects to a rework bench (bypass lane) and scraps repeat offenders. Finished bikes are packed in batches of 3. 'cnc_line' is a user-defined node type built from a template.

Generated 2026-10-13 19:48 | simulated 2.0 h | seed 7 | 10 replications

HEALTH SCORE

78 /100

Grade C

Acceptable: workable, but there are clear risks to address before scaling up.

Critical 0 | High 1 | Medium 3 | Low 1 | Info 1

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
285.0 units/h	570 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
24.6 units	283 s	paint

Executive summary

Primary bottleneck: 'paint' (87% busy, 10% down, index 100). Sources are held back up to 26% of the time. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (paint): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'paint' (+8.0% throughput).
- Medium (supply): The line cannot absorb the offered supply, so everything upstream of the constraint spends time waiting.
- Medium (paint): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.
- Medium (paint): Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.

Best measured improvement

Best lever: 10% faster processing at 'paint' -> +24.8 units/hour (8.0%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**paint**	The bottleneck is 98% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'paint' (+8.0% throughput).	-8
Medium	Flow	**supply**	Upstream supply is held back up to 26% of the time by a full line. Action: The line cannot absorb the offered supply, so everything upstream of the constraint spends time waiting.	-5
Medium	Reliability	**paint**	Down 10% of the time (5 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Medium	Reliability	**paint**	Every unit passes through this station, and it can fail. A single failure stops the whole plant. Action: Add a parallel machine or a buffer of finished work in front of the downstream stations so they can ride through a repair.	-4
Low	Model	**run length**	The measured window (2.0 h) is only 25x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**paint**	Constraint of the line: 98% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'paint' (+8.0% throughput).	

Performance by node

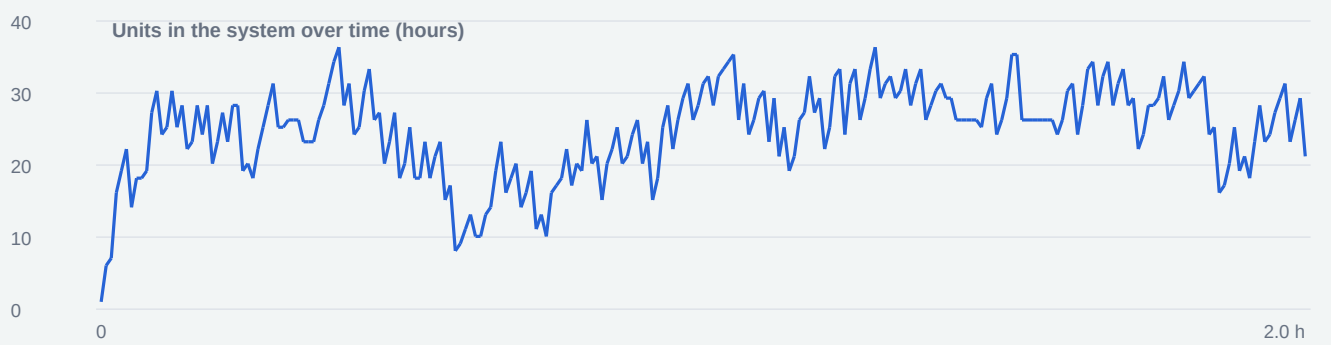
Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
frames	source	-	-	-	-	-
wheels	source	-	-	-	-	-
paint bottleneck	machine		87.4	0.0	10.4	4.0 / 5
wheel_belt	delay		8.9	81.1	0.0	7.2 / 12
kitter	assembler		73.9	0.0	0.0	6.0 / 9
qc	quality_gate		32.0	0.0	0.0	1.2 / 9
bench	machine		46.6	0.0	0.0	1.7 / 9
pack	batcher		13.8	0.0	0.0	4.1 / 15

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
ship	sink	-	-	-	-	-
bin	sink	-	-	-	-	-

Work in process over time



Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	299.10 units/h	+- 10.50	14.68	3.5%
Average WIP	26.24 units	+- 0.70	0.97	2.6%
Cycle time	295.15 s	+- 9.62	13.45	3.3%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
paint	20 min	2.0 min	90.9%	10.4%	5

Single points of failure

All units must pass through: paint, kitter, qc. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 635 = completed 570 + scrapped 42 + lost 0 + in system 23
Little's Law	PASS	L = 24.60, lambda x W = 24.08 (gap 2.1%)
Run length	WARN	window = 25 x cycle time (30x or more recommended)
Warm-up	WARN	no warm-up: start-up transients are included in averages
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 308.3 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
paint	10% faster processing	333.1	**+24.8**	+8.0%
paint	+2 buffer slots	311.5	**+3.2**	+1.0%
paint	20% shorter repairs	311.3	**+3.1**	+1.0%
kitter	+2 buffer slots	308.5	**+0.3**	+0.1%
kitter	10% faster processing	308.3	**+0.0**	+0.0%
bench	+2 buffer slots	308.3	**+0.0**	+0.0%
bench	10% faster processing	308.3	**+0.0**	+0.0%

Best lever: 10% faster processing at 'paint' -> +24.8 units/hour (8.0%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
frames	source	interarrival=exponential(26), mix={"frame":1}, on_blocked=wait
wheels	source	interarrival=exponential(13), mix={"wheel":1}, on_blocked=wait
paint	cnc_line based on machine	capacity=4, mtbf=1200, mttr=120, on_failure=restart, process=normal(30,5)
wheel_belt	delay	capacity=12, transit=uniform(12,6)
kitter	assembler	capacity=4, kit={"frame":1,"wheel":2}, output_type=bike, process=normal(26,4)
qc	quality_gate	capacity=3, max_reworks=1, pass_to=pack, process=exponential(8), rework_to=bench, scrap_to=bin, threshold=0.14
bench	machine	capacity=3, process=normal(45,8)
pack	batcher	batch_size=3, capacity=6, process=constant(15), timeout=150
ship	sink	counts_as=completed
bin	sink	counts_as=scrapped

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**paint**	The bottleneck is 98% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**supply**	Upstream supply is held back up to 26% of the time by a full line.	-5
Medium	**paint**	Down 10% of the time (5 failures).	-3
Medium	**paint**	Every unit passes through this station, and it can fail. A single failure stops the whole plant.	-4
Low	**run length**	The measured window (2.0 h) is only 25x the cycle time.	-2
	Total deductions		**-22**
	Health score		**78**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.